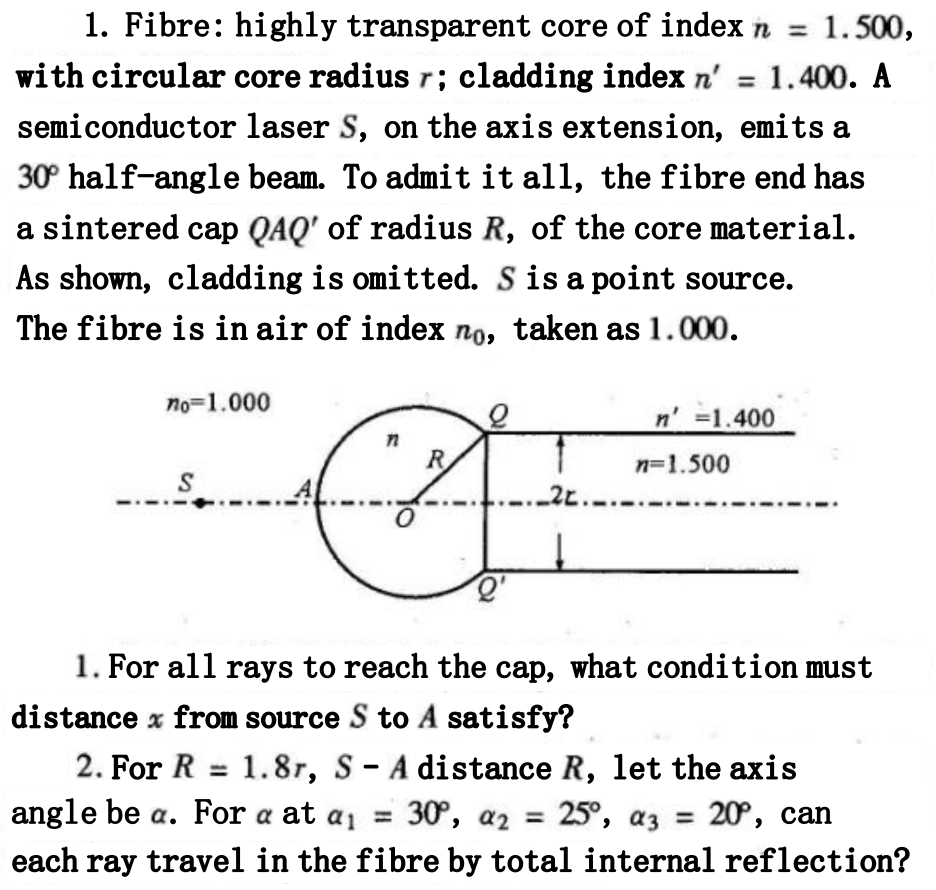
21st National High School Physics Competition Final Problems

1. (20 points)

Highly transparent fibre core: $n=1.500$, radius r; cladding index $n'=1.400$. Semiconductor laser S, on the axis extension, emits a beam of half-angle 30°. To admit all the light, a spherical cap QAQ′ of radius R, of the core material, is sintered at its end.
The figure omits the cladding. S is a point source; the fibre is in air, with index n_0 taken as 1.000.
1. For the whole beam to reach the cap, what condition must distance x from source S to A satisfy?
2. For $R=1.8r$, S - A separation R, axis angle α: for α at $\alpha_1=30°$、$\alpha_2=25°$ and $\alpha_3=20°$, can each of the three rays propagate in the fibre by total internal reflection?



1. Fibre: highly transparent core of index $n = 1.500$, with circular core radius r; cladding index $n' = 1.400$. A semiconductor laser S, on the axis extension, emits a 30° half-angle beam. To admit it all, the fibre end has a sintered cap QAQ' of radius R, of the core material. As shown, cladding is omitted. S is a point source. The fibre is in air of index n_0, taken as 1.000.

n_0=1.000
Q
n' =1.400
n
R
n=1.500
S
A
2r
O
Q′

1. For all rays to reach the cap, what condition must distance x from source S to A satisfy?

2. For $R = 1.8r$, S - A distance R, let the axis angle be α. For α at $\alpha_1 = 30°$, $\alpha_2 = 25°$, $\alpha_3 = 20°$, can each ray travel in the fibre by total internal reflection?

2. (20 points)

Use relativistic energy and momentum to analyse and justify the following.
1. A photon colliding with a free electron at rest in vacuum cannot have its energy fully absorbed.
2. A photon on a metal can be fully absorbed, ejecting an electron by the photoelectric effect.

3. (25 points)

Shown wooden wedge OABC ($M=30.0\text{kg}$) rests on level ground; AB has slope $\beta=60°$, and BC slope $\alpha=45°$. A very short smooth arc joins AB, BC at B. Release a point block, $m=4.00\text{kg}$, from rest at A, height $h_1=2.80\text{m}$ above ground. It slides without friction on the wedge.
Maximum static and sliding ground friction have coefficients $\mu=6.00\times10^{-2}$. The height of B above the table is $h_2=2.00\text{m}$. Ignore both speed changes at B. Find work done on the wedge from A to the bottom C by the sliding block. Take $g=10.0\text{m/s}^2$.